10.6 Segment Relationships in Circles

Name _____ Period ____ Date _____

Geometry 10.6 · Student sheet

I can…

- find lengths of segments of chords that cross inside a circle.
- find lengths of secant segments that meet outside a circle.
- find lengths when a tangent and a secant meet outside a circle.

Vocabulary · fill in the missing word

Segments of a chord — the two _____ of a chord that is cut by another chord

Tangent segment — a segment that is tangent to a circle at one of its _____

Secant segment — a segment that _____ a chord and has exactly one endpoint outside the circle

External segment — the part of a secant segment that is _____ the circle

Part 1 · Discovery: crossing chords

Draw a large circle and two chords $\overline{AB}$ and $\overline{CD}$ that cross at E. Measure the four pieces in millimeters.

EA	EB	EA · EB	EC	ED	EC · ED
Partner:					

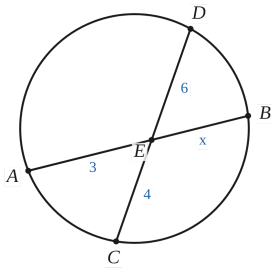
What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

Segments of Chords Theorem: if two chords cross inside a circle, the _____ of the lengths of the segments of one chord equals the product for the other chord: EA · EB = EC · ED.

Example 1

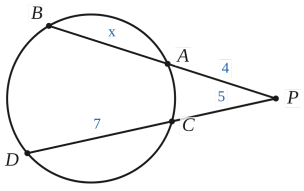
Chords $\overline{AB}$ and $\overline{CD}$ cross at E. Find the value of x.



Segments of Secants Theorem: if two secants share an endpoint outside a circle, then (external segment) · (_____) is the same for both.

Example 2

Find the value of x.



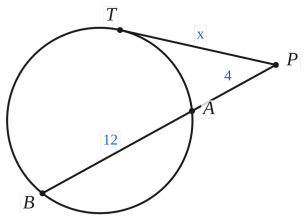
Example 3

Two chords cross. The pieces of one chord are x and x + 4. The pieces of the other chord are 3 and 4. Find x.

Segments of Secants and Tangents Theorem: if a tangent and a secant share an endpoint outside a circle, then the _____ of the tangent segment = (external segment) · (whole secant segment).

Example 4

$\overline{PT}$ is tangent to the circle at T. Find the value of x.



Example 5

You stand 8 feet from the edge of a round silo. The distance from you to a point of tangency on the silo is 16 feet. What is the radius of the silo?

Part 3 · Partner practice

Partner 1

Two chords cross. The pieces of one are 5 and x; the pieces of the other are 10 and 4. Find x.

Partner 2

A tangent segment is 12. A secant from the same point has an external segment of 8 and an internal segment of x. Find x.